

CHRISTIAN MILLAR

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SUMMARY

Hands-on systems engineer with professional experience in electromechanical hardware integration, embedded control systems, and industrial fieldbus protocols. Leads root-cause anomaly investigations spanning electrical, mechanical, firmware, and protocol layers. Builds test automation and diagnostic tooling in Python. Background in autonomous systems, robotic mechanisms, and solid rocket propulsion test campaigns.

PROFESSIONAL EXPERIENCE

Systems Integration & Test Engineer

2025 – Present

Advanced Motion Controls

Camarillo, CA

- Serve as primary technical authority for full servo drive product portfolio (FlexPro, DigiFlex Performance, AxCent), owning end-to-end system integration, hardware bring-up, closed-loop control tuning, and multi-protocol communication validation across CANopen, EtherCAT, EtherNet/IP, and Modbus RTU for customers in medical, defense, and industrial automation sectors
- Lead hands-on root-cause investigations on fielded multi-axis electromechanical systems: diagnosed resolver feedback errors from a transformer ratio mismatch, resolved Hall and encoder commutation faults on legacy brushless motors, traced current commands exceeding configured limits during mode transitions, and isolated EtherCAT state machine failures caused by PDO mapping mismatches
- Architect and develop Python-based diagnostic tooling for real-time fieldbus analysis and automated data review; built custom CAN bus decoder capable of parsing multi-node traffic against the full CANopen DS301/DS402 object dictionary, enabling rapid fault isolation across networked drive systems
- Perform on-site system setup, commissioning, and integration at customer facilities; configure servo drives, motors, feedback devices, fieldbus communications, and control parameters, then troubleshoot electrical, mechanical, firmware, and network-level issues to achieve reliable closed-loop operation
- Drive cross-functional failure investigations with hardware and firmware engineering; independently identify discrepancies between product documentation and protocol specifications (CiA 402), author root-cause reports, test procedures, and corrective action recommendations that feed directly into firmware and documentation updates
- Own technical documentation for the product line: author integration procedures, commissioning guides for multi-axis systems (up to 4-axis coordinated configurations), protocol command references, and hardware configuration matrices spanning 59+ active product models and 300+ total products

Project Lead – Autonomous Systems Integration & Test

2024 – 2025

University of California, Los Angeles

Los Angeles, CA

- Led 5-engineer team through full systems engineering lifecycle for autonomous navigation platform: defined requirements, designed system architecture, and integrated infrared, ultrasonic, and PID-controlled servo subsystems in MATLAB/Simulink Stateflow; designed modular test harness with automated fault detection, reducing integration debug time by 50%

Lead Mechanical Engineer – Robotic Manipulator

2023

University of California, Los Angeles

Los Angeles, CA

- Designed 3-DOF robotic manipulator for aerospace/defense applications; performed structural analysis (FEA, hand calculations) to validate dynamic load integrity; produced 100+ engineering drawings with full GD&T and managed configuration control through iterative design cycles

TECHNICAL PROJECTS

Solid Rocket Motor Development & Static-Fire Test Campaign

2017 – 2019

- Designed, fabricated, and tested M-class composite solid rocket motor through two instrumented static-fire test campaigns. Measured 2,081 lbf·s total impulse, 948 lbf peak thrust, 1,028 psi chamber pressure, 237.5 s Isp

Remotely Operated Underwater Vehicle (ROV)

2025 – Present

- Designed and built tethered ROV with Arduino-based embedded control, custom propulsion, and engineered pumped-fluid-loop active buoyancy control system with calculated pressure drops and flow modeling

EDUCATION

B.S. Mechanical Engineering | GPA: 3.48

2020 – 2024

University of California, Los Angeles

Los Angeles, CA

Key Coursework: Robotic Controls, Autonomous Systems, Computational Electromagnetics, Electronic Circuit Design, Embedded Systems, Control Systems

TECHNICAL SKILLS

Protocols: CANopen (DS301/DS402), EtherCAT (CoE), EtherNet/IP, Modbus RTU, RS-485/232, JTAG/Boundary Scan, SPI, I2C

Software: Python (test automation, data analysis), C++, JavaScript/TypeScript, React, MATLAB/Simulink, Git, Linux

Test & Hardware: Oscilloscopes, protocol analyzers, boundary scan, servo commissioning, EMC diagnostics, power electronics

CAD & Design: SolidWorks, Fusion 360, FEA, GD&T | Servo control, motor commutation, feedback devices, PID tuning